

SIALKOT, Pakistan

WMO: 415990

Lat: 32.500N

Lon: 74.533E

Elev: 837

StdP: 14.26

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 39.0	(c) 41.0	(d) 34.0	(e) 29.5	(f) 47.8	(g) 36.8	(h) 33.0	(i) 49.8	(j) 13.1	(k) 57.5	(l) 11.3	(m) 57.2	(n) 0.9	(o) 310	(p) 0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 22.6	(c) 107.9	(d) 72.4	(e) 105.1	(f) 72.8	(g) 102.1	(h) 72.7	(i) 84.0	(j) 92.8	(k) 83.2	(l) 91.9	(m) 82.4	(n) 91.1	(o) 5.7	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.0	171.9	89.0	81.0	166.0	88.9	80.2	162.0	88.6	49.2	92.4	48.3	92.0	47.4	90.6	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Extreme Max WB
(a) 16.6	(b) 12.3	(c) 10.3	(d) DB	(e) 34.9	(f) 112.8	(g) 1.0	(h) 2.1	(i) 34.2	(j) 114.3	(k) 33.6	(l) 115.6	(m) 33.1	(n) 116.7	(o) 32.4	(p) 118.2	
(a) 16.6	(b) 12.3	(c) 10.3	(d) WB	(e) 34.3	(f) 86.2	(g) 1.9	(h) 0.9	(i) 32.9	(j) 86.8	(k) 31.8	(l) 87.3	(m) 30.8	(n) 87.8	(o) 29.4	(p) 88.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.1	52.8	59.2	67.8	79.1	87.3	89.9	86.7	85.3	83.2	76.3	65.2	55.9
	DBStd	13.53	4.17	4.47	5.78	5.65	4.60	5.01	4.07	3.30	3.24	4.59	3.86	4.41
	HDD50	26	19	1	0	0	0	0	0	0	0	0	0	6
	HDD65	916	379	170	37	0	0	0	0	0	0	0	46	284
	CDD50	8830	105	258	553	872	1156	1198	1139	1095	997	815	455	187
	CDD65	4246	0	7	125	422	691	748	674	630	547	350	51	1
	CDH74	56704	2	67	1224	5179	10327	11658	9552	8435	6518	3287	435	20
CDH80	31517	0	1	402	2894	6602	7619	5324	4271	3031	1325	47	1	
Wind	WSAvg	3.4	2.6	3.4	4.0	4.7	5.0	4.9	3.8	3.1	3.0	2.4	1.9	2.2
Precipitation	PrecAvg	38.2	1.5	1.5	1.7	0.9	0.7	3.0	10.8	10.0	4.8	0.5	0.4	0.8
	PrecMax	74.3	4.5	5.4	5.8	2.8	1.8	6.4	27.3	36.1	18.9	1.9	1.7	4.4
	PrecMin	21.8	0.0	0.0	0.0	0.0	0.0	0.4	1.5	3.1	0.7	0.0	0.0	0.0
	PrecStd	15.9	1.4	1.3	1.6	0.7	0.6	1.8	7.0	8.4	4.2	0.6	0.5	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.2	79.0	93.3	104.4	111.4	111.6	105.5	97.1	96.4	93.4	83.8	76.6
		MCWB	58.4	62.9	68.7	71.3	70.3	72.9	78.6	82.5	80.3	77.6	67.2	60.9
	2%	DB	68.3	76.0	87.9	100.8	107.7	109.1	100.5	95.4	94.5	91.4	81.3	73.4
		MCWB	56.7	61.5	68.5	69.9	70.8	73.1	79.8	82.0	79.9	75.5	66.7	59.9
	5%	DB	66.1	73.3	84.3	98.2	104.4	106.1	97.5	94.6	93.1	89.6	79.0	70.1
		MCWB	55.0	60.1	66.2	68.9	70.5	73.6	80.8	81.8	79.8	74.3	65.3	58.2
	10%	DB	63.0	70.1	80.7	93.8	101.8	103.4	95.1	93.0	91.3	87.4	76.6	67.6
		MCWB	54.0	58.6	65.0	68.1	70.6	74.0	80.6	81.7	78.7	72.5	63.5	57.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.1	65.4	73.9	76.0	78.9	83.3	85.1	85.2	83.9	80.8	71.2	64.1
		MCDB	68.5	75.6	86.2	97.1	99.0	95.7	94.6	92.8	92.7	89.5	80.3	72.2
	2%	WB	58.2	63.2	70.4	73.2	76.1	81.9	84.0	84.1	82.4	78.8	68.7	61.2
		MCDB	65.3	73.4	84.3	91.6	97.2	93.7	93.4	92.1	90.7	87.8	78.2	70.3
	5%	WB	56.4	61.5	68.0	71.2	74.2	80.4	83.1	83.3	81.2	77.2	66.8	59.4
		MCDB	63.6	70.8	80.5	91.4	96.0	93.6	92.2	91.4	89.8	85.6	77.0	67.8
	10%	WB	54.8	59.9	66.0	69.9	72.7	79.0	82.3	82.5	80.1	75.4	64.8	57.6
		MCDB	61.5	68.5	78.3	88.7	94.1	92.9	91.2	90.6	88.6	84.6	74.2	66.3
Mean Daily Temperature Range	5% DB	MDBR	19.8	20.6	22.7	26.2	26.4	22.6	15.0	13.4	15.7	21.5	23.5	22.6
		MCDBR	24.7	24.6	27.1	31.1	30.7	28.1	18.3	15.6	17.1	21.4	24.9	26.9
		MCWBR	14.3	12.6	11.8	10.5	10.0	9.4	7.4	7.0	7.6	9.0	12.8	15.5
	5% WB	MCDBR	21.7	21.9	25.0	27.5	26.8	20.0	15.5	14.1	15.5	18.3	23.1	25.3
		MCWBR	13.6	12.5	12.2	11.2	10.9	8.7	7.4	7.2	7.5	8.5	13.6	15.7
Clear-Sky Solar Irradiance	taub		0.461	0.490	0.499	0.542	0.596	0.686	0.734	0.665	0.574	0.571	0.529	0.481
	taud		1.921	1.901	1.910	1.855	1.756	1.623	1.584	1.717	1.862	1.759	1.785	1.884
	Ebn at Noon		227	236	247	243	231	210	199	211	224	209	203	211
	Edh at Noon		49	56	59	65	72	82	85	74	61	63	56	49
All-Sky Solar Radiation	RadAvg		940	1223	1654	1996	2269	2208	1826	1735	1728	1564	1191	977
	RadStd		102	151	133	99	75	122	152	131	83	76	53	61

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page